

SynapseLab

One-Page Concept Summary — DataForge 2026

Problem Statement

Learning and memory are often described using abstract neuroscience terms such as synaptic plasticity, strengthening of connections, and interference. These ideas are difficult for beginners to visualize, while many AI systems hide their internal representations. SynapseLab addresses this gap by providing a small, transparent interactive environment where users can observe a simplified memory model learn patterns and then examine how interference affects recall.

Proposed Solution

SynapseLab is a browser-based computational simulation inspired by the idea that changes in connection strength can contribute to learning and memory. Rather than claiming to reproduce the biological brain, it creates an interpretable artificial memory system. Users can train the system on simple patterns, recall them, introduce controlled interference, and compare the resulting predictions. This makes the relationship between learning, stored weights and retrieval directly observable.

How the Experiment Works

The experiment uses four easy-to-understand symbolic classes: CAT, DOG, BIRD and CAR. A selected pattern is presented during learning, and the model updates its internal representation. During recall, the input is compared with the learned representations and the system predicts the most likely class. Users can then apply a 40% interference setting. In the simulation, this means that a substantial portion of the relevant representation is deliberately perturbed or mixed with competing information, allowing users to observe reduced confidence or altered recall.

Neuroscience Motivation

The concept is motivated by research on synaptic plasticity and memory. Biological synapses can change their strength in response to activity, and such changes are associated with learning and memory processes. SynapseLab translates this broad scientific idea into an intentionally simplified computational demonstration. The 40% interference value and the CAT/DOG/BIRD/CAR task are project-specific simulation choices, not measurements of human memory or biological synaptic change.

Technology and Innovation

The application is built with Next.js and TypeScript, with the model logic separated into reusable TypeScript modules. It is deployed as a web application and its source code is publicly available on GitHub. Its key innovation is accessibility and transparency: instead of presenting only an accuracy score, SynapseLab turns a memory-like learning process into an interactive visual experiment. The design can support education in AI, computational neuroscience and explainable machine learning.

Impact, Limitations and Future Scope

SynapseLab can help students understand how learned representations may be affected by competing information and can provide a foundation for classroom demonstrations. However, it is not a biological brain model, clinical tool, or validated cognitive assessment. Future versions could add adjustable learning rates, multiple interference levels, noise controls, weight visualizations, recall-over-time graphs, larger pattern sets, and quantitative comparisons between different learning strategies. These additions would enable more systematic computational experiments while preserving the project's emphasis on interpretability.

Project Links

Live Demo: <https://synapse-lab-three.vercel.app> **GitHub:** <https://github.com/sagarpandare21/SynapseLab>

Scientific note: SynapseLab distinguishes established neuroscience motivation from its own simulation assumptions. It is intended as an educational computational model, not evidence that the exact simulated behavior occurs in the human brain.

Selected reference: Huang M. et al. (2023), iScience, 26(5), 106664, PMID 37168570, DOI 10.1016/j.isci.2023.106664.